

Practical -2 _OS_2520090235

```
GNU nano 8.7.1
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/types.h>
#include <sys/wait.h>
#include <string.h>

int main() {

    char command[100];

    // Accept Linux command from the user
    printf("Enter a Linux command: ");
    fgets(command, sizeof(command), stdin);

    // Remove newline character
    command[strcspn(command, "\n")] = '\0';

    // Create child process
    pid_t pid = fork();

    if (pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    // Child Process
    else if (pid == 0) {

        printf("\nChild Process\n");
        printf("Child PID : %d\n", getpid());
        printf("Parent PID : %d\n", getppid());

        printf("\nExecuting command: %s\n\n", command);

        // Execute Linux command
        execlp(command, command, NULL);

        // Executes only if exec fails
        printf("Command execution failed!\n");
        exit(1);
    }

    // Parent Process
    else {

        printf("\nParent Process\n");
        printf("Parent PID : %d\n", getpid());
        printf("Child PID : %d\n", pid);

        // Wait for child process
        wait(NULL);

        printf("\nChild process completed.\n");
    }

    return 0;
}
```

TASK 1

OUTPUT:

```

rohita@LaxmiDurga:~$ ./a.out
Enter a Linux command: ls

Parent Process
Parent PID : 1240
Child PID : 1241

Child Process
Child PID : 1241
Parent PID : 1240

Executing command: ls

First.c Hello-World Ossp Practical2.c Skill2.c Skill3.c Skill3Task2.c a.out command_exec.c nano.565.save ossp skil4Task1.c skill3Task2.c
Child process completed.

```

TASK -2

Uname -a : Identifies the core system kernel and hardware architecture.

```

Parent Process
Parent PID : 1240
Child PID : 1241

Child Process
Child PID : 1241
Parent PID : 1240

Executing command: ls

First.c Hello-World Ossp Practical2.c Skill2.c Skill3.c Skill3Task2.c a.out command_exec.c nano.565.save ossp skil4Task1.c skill3Task2.c
Child process completed.
rohita@LaxmiDurga:~$ nano Practical2.c
rohita@LaxmiDurga:~$ uname -a
Linux LaxmiDurga 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2026 x86_64 GNU/Linux
rohita@LaxmiDurga:~$ lscpu
Architecture:                x86_64
CPU op-mode(s):              32-bit, 64-bit
Address sizes:                39 bits physical, 48 bits virtual
Byte Order:                   Little Endian
CPU(s):                       4
On-line CPU(s) list:         0-3
Vendor ID:                    GenuineIntel
Model name:                   Intel(R) Core(TM) i3-5010U CPU @ 2.10GHz
CPU family:                   6
Model:                        61
Thread(s) per core:          2
Core(s) per socket:          2
Socket(s):                    1
Stepping:                     4
BogoMIPS:                     4190.30
Flags:                        fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm arch
2 movbe popcnt aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch pti ssbd ibrs ibpb stibp fsgsbase bmi1 avx2 smep bmi2 erms
Virtualization features:
Hypervisor vendor:           Microsoft
Virtualization type:         full
Caches (sum of all):
L1d:                          64 KiB (2 instances)
L1i:                          64 KiB (2 instances)
L2:                           512 KiB (2 instances)
L3:                           3 MiB (1 instance)
NUMA:
NUMA node(s):                 1
NUMA node0 CPU(s):           0-3
Vulnerabilities:
Gather data sampling:         Not affected
Ghostwrite:                   Not affected
Indirect target selection:     Mitigation; Aligned branch/return thunks

```

Lscpu : Details the physical CPU geometry (sockets, cores, hardware caches).

```

rohita@LaxmiDurga:~$ ps aux
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.2  0.3 24376 15388 ?        Ss   16:08   0:03 /usr/lib/systemd/systemd --system --deserialize=79
root         2  0.0  0.0   3180  2208 hvc0?    Sl+  16:08   0:00 /init
root         8  0.0  0.0   3180  2048 hvc0?    Sl+  16:08   0:00 plan9 --control-socket 8 --log-level 4 --server-fd 9 --pipe-fd 11 --log-truncate
root       121  0.0  0.0   3184  1112 ?        Ss   16:08   0:00 /init
root       122  0.0  0.0   3280  1252 ?        S    16:08   0:00 /init
rohita     123  0.0  0.1  6288  5544 pts/0    Ss   16:08   0:00 -bash
root       355  0.0  0.0   2888  1944 ?        Ss   16:09   0:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root       356  0.0  0.0   4472  3080 ?        Ss   16:09   0:00 /usr/sbin/cron -f -P
message+   358  0.0  0.1   9012  5652 ?        Ss   16:09   0:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --systemd-activation --syslog-only
root       365  0.0  0.7 44096 29840 ?        Ss   16:09   0:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root       370  0.0  0.2  17788  8792 ?        Ss   16:09   0:00 /usr/lib/systemd/systemd-logind
root       390  0.0  0.0   5492  2756 ttyl?    Ss+  16:09   0:00 /usr/sbin/agetty --noneset --noclear --issue-file=/etc/issue:/etc/issue.d/run/issue.d:/usr/lib/issue.d - linux
syslog     412  0.0  0.1  220548  5664 ?        Ssl  16:09   0:00 /usr/sbin/rsyslogd -n -iNONE
root       415  0.0  0.8 123460 32644 ?        Ssl  16:09   0:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-shutdown --wait-for-signal
_chrony    434  0.0  0.2  24036 11492 ?        S    16:09   0:00 /usr/sbin/chronyd -n -F 1 -x
_chrony    464  0.0  0.0  12096  2340 ?        S    16:10   0:00 /usr/sbin/chronyd -n -F 1 -x
root       916  0.0  0.3  34088 12668 ?        Ss   16:14   0:00 /usr/lib/systemd/systemd-journald
root       976  0.0  0.2  33776 10792 ?        Ss   16:14   0:00 /usr/lib/systemd/systemd-udev
systemd+  1037  0.0  0.3  22416 14472 ?        Ss   16:15   0:00 /usr/lib/systemd/systemd-resolved
polkitd    1153  0.0  0.2  306444  8636 ?        Ssl  16:17   0:00 /usr/lib/polkit-1/polkitd --no-debug --log-level=notice
rohita    1284  0.0  0.1   7160  4428 pts/0    R+   16:32   0:00 ps aux

```

Ps aux: Gives a idea of running software processes and their resource states

Top: Shows a real-time, dynamic view of active CPU and Virtual Memory/Swap consumption.

```
top - 16:34:20 up 26 min,  0 users,  load average: 0.00, 0.05, 0.26
Tasks: 21 total,  1 running, 20 sleeping,  0 stopped,  0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :  3876.6 total,  3350.4 free,   486.7 used,   181.3 buff/cache
MiB Swap:  1024.0 total,  1024.0 free,    0.0 used.  3389.9 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
1	root	20	0	24376	15388	11324	S	0.0	0.4	0:03.35	systemd
2	root	20	0	3180	2208	2072	S	0.0	0.1	0:00.16	init-systemd(Ub
8	root	20	0	3180	2048	1940	S	0.0	0.1	0:00.00	init
121	root	20	0	3184	1112	980	S	0.0	0.0	0:00.00	SessionLeader
122	root	20	0	3200	1252	1108	S	0.0	0.0	0:00.02	Relay(123)
123	rohita	20	0	6268	5544	3864	S	0.0	0.1	0:00.16	bash
355	root	20	0	2888	1944	1820	S	0.0	0.0	0:00.15	chronyd-starter
356	root	20	0	4472	3080	2832	S	0.0	0.1	0:00.02	cron
358	message+	20	0	9012	5652	4684	S	0.0	0.1	0:00.49	dbus-daemon
365	root	20	0	44096	29840	14696	S	0.0	0.8	0:00.30	networkd-dispat
370	root	20	0	17788	8792	7580	S	0.0	0.2	0:00.27	systemd-logind
390	root	20	0	5492	2756	2572	S	0.0	0.1	0:00.01	agetty
412	syslog	20	0	220548	5664	4364	S	0.0	0.1	0:00.08	rsyslogd
415	root	20	0	123460	32644	18036	S	0.0	0.8	0:00.23	unattended-upgr
434	_chrony	20	0	24036	11492	7132	S	0.0	0.3	0:00.17	chronyd
464	_chrony	20	0	12096	2340	1696	S	0.0	0.1	0:00.00	chronyd
916	root	19	-1	34088	12668	11492	S	0.0	0.3	0:00.06	systemd-journal
976	root	20	0	33776	10792	9108	S	0.0	0.3	0:00.10	systemd-udev
1037	systemd+	20	0	22416	14472	11980	S	0.0	0.4	0:00.10	systemd-resolve
1153	polkitd	20	0	306444	8636	7820	S	0.0	0.2	0:00.15	polkitd
1294	rohita	20	0	8276	6036	3908	R	0.0	0.2	0:00.00	top